TECHNIQUE MASTERING EXERCISES

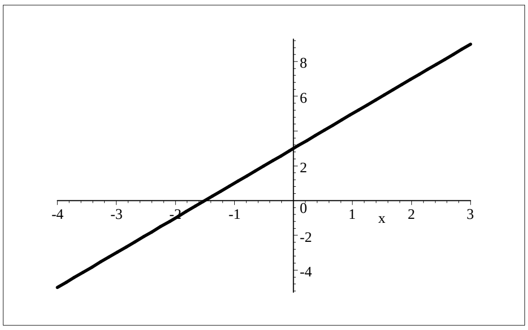
F - GRAPHS

SECTIONS C GRAPHS

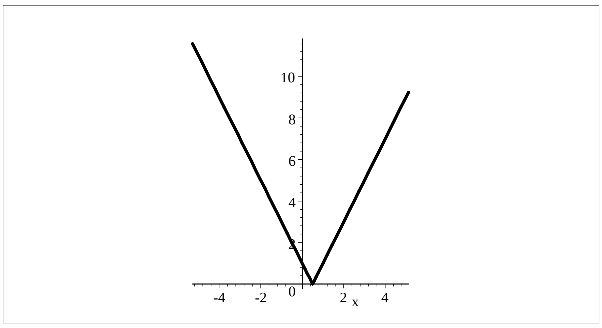
1.

a. $f(x) = 2x + 3$ on $[-4,\ 3]$

b. $f(x) = |2x-1| = \begin{cases} 2x-1 & \text{if } \quad x \geq \dfrac{1}{2} \\ -2x+1 & \text{if } \quad x < \dfrac{1}{2} \end{cases}$



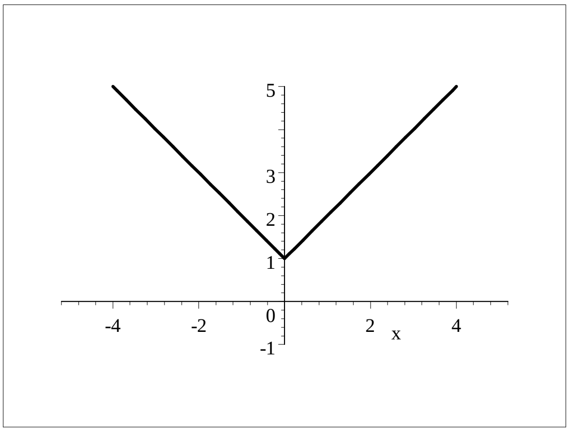
a.



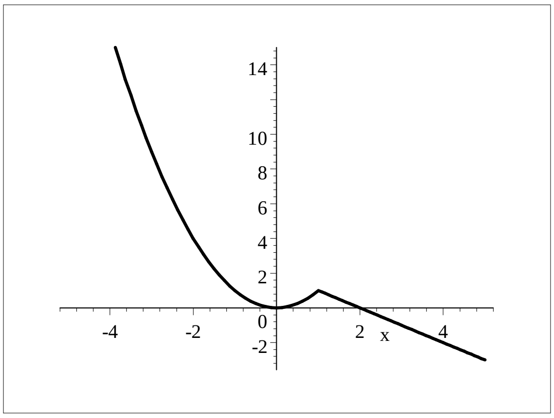
b.

c. $f(x) = |x| + 1$

d. $f(x) = \begin{cases} x^2 & \text{if } \quad x \leq 1 \\ -x+2 & \text{if } \quad x > 1 \end{cases}$



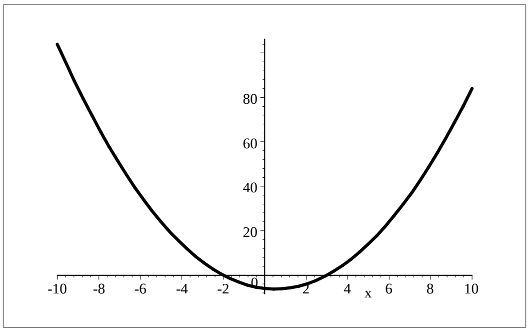
c.



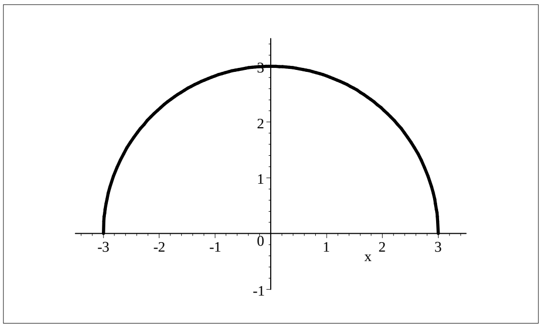
d.

e. $f(x) = x^2 - x - 6 = (x+2)(x-3)$

f. $f(x) = \sqrt{9-x^2} \quad x \in [-3,\ 3]$

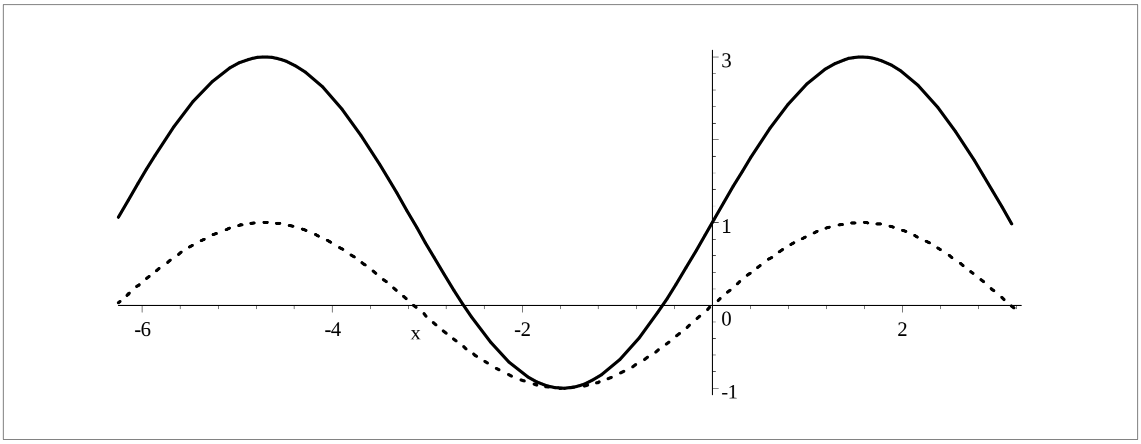


e.



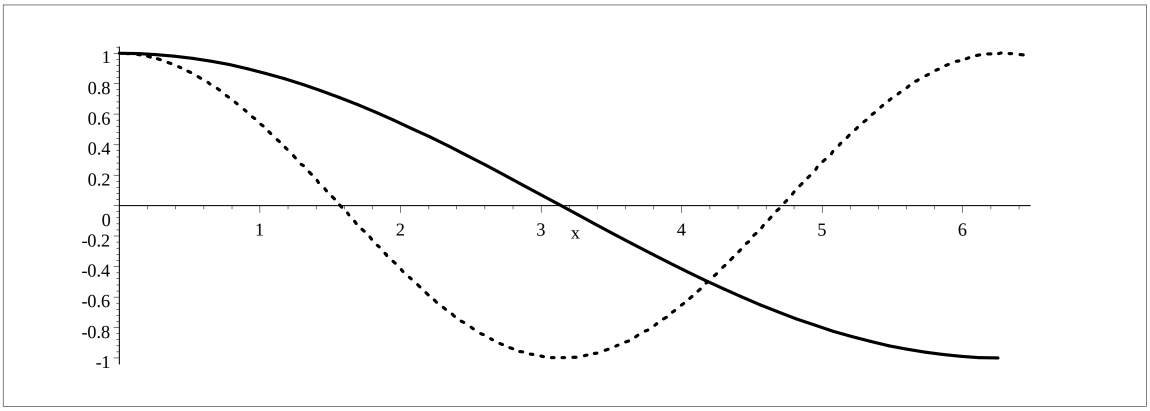
f.

g. $f(x) = 2\sin x + 1$ on $[-2\pi,\ \pi]$,



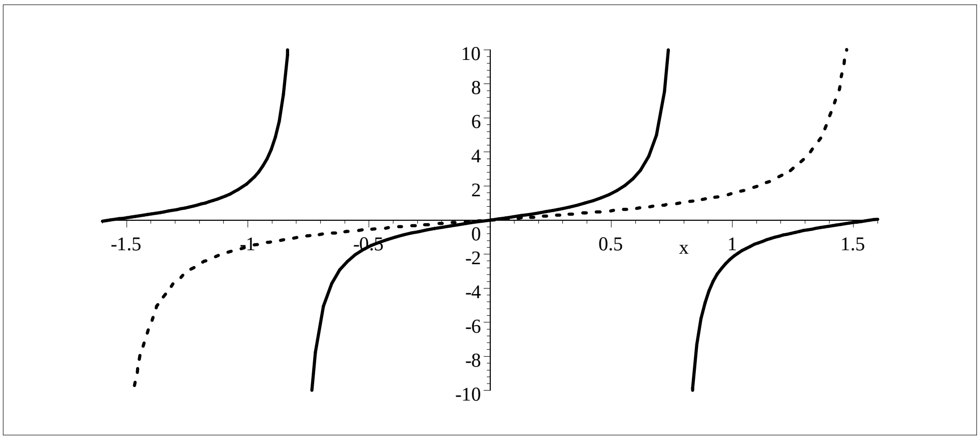
$$\left[f \ \underline{\qquad},\ y = \sin x \ \ \ldots\right]$$

h. $f(x) = \cos\frac{x}{2}$ on $[0,\ 2\pi]$,



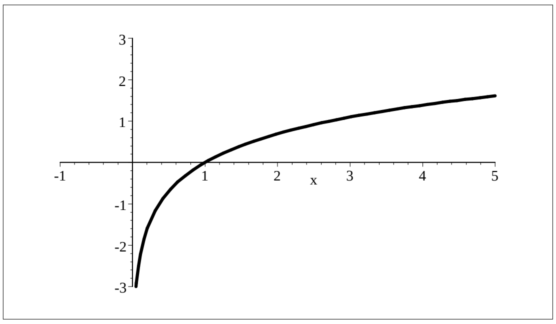
$$\left[f \ \underline{\qquad},\ y = \cos x \ \ \ldots\right]$$

i. $f(x) = \tan 2x$ on $\left[\frac{-\pi}{2},\ \ \frac{\pi}{2}\right]$

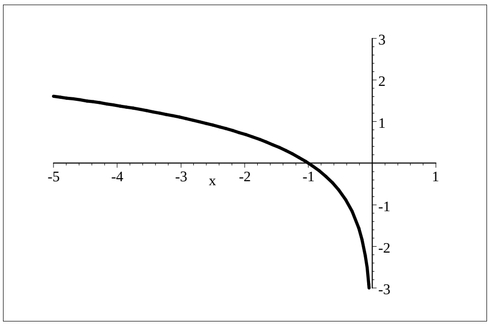


$$\left[f \ \underline{\qquad},\ y = \tan x \ \ \ldots\right]$$

2. **a.** $f(x) = \ln x, D_f = (0,\infty), W_f = \mathbb{R}$. **b.** $g(x) = \ln(-x), D_g = (-\infty,0), W_g = \mathbb{R}$.



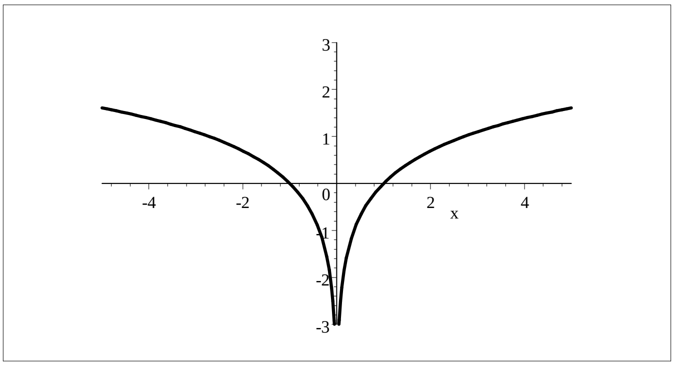
a.



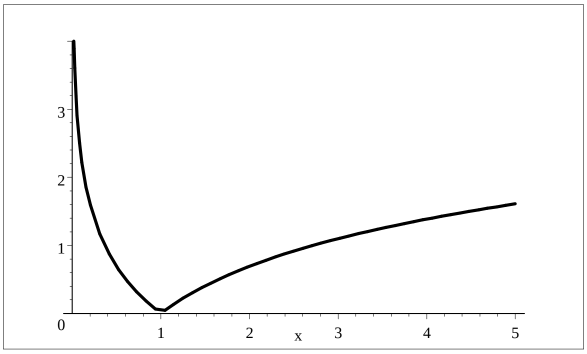
b.

c. $h(x) = \ln|x| = \begin{cases} \ln x & \text{if } x > 0 \\ \ln(-x) & \text{if } x < 0 \end{cases}$ $D_h = \mathbb{R}\backslash\{0\}, W_h = \mathbb{R}$.

d. $l(x) = |\ln x| = \begin{cases} \ln x & \text{if } x \geq 1 \\ -\ln x & \text{if } 0 < x < 1 \end{cases}$ $, D_l = (0,\infty),\ W_l = [0,\infty)$.

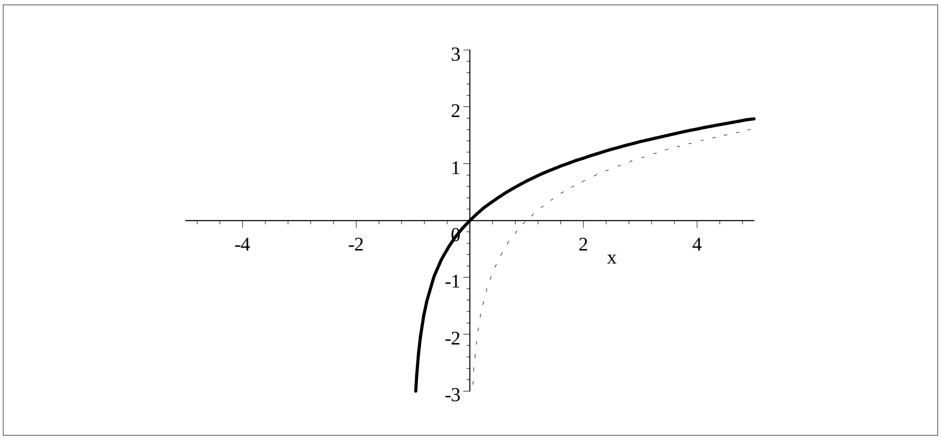


c.



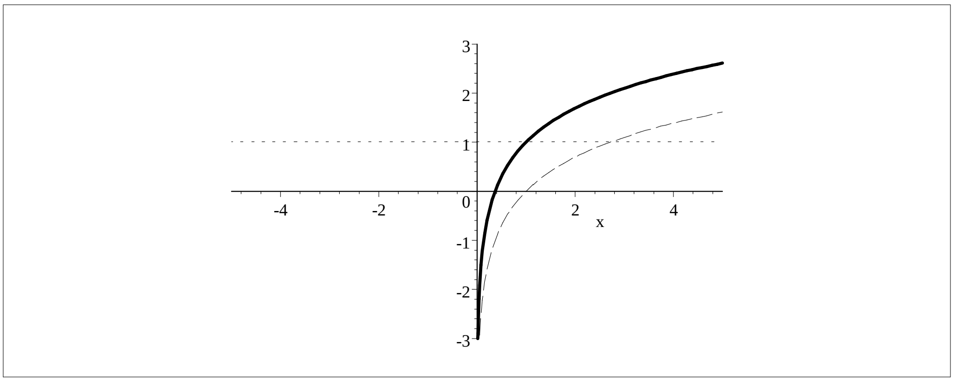
d.

e. $m(x) = \ln(1+x), D_m = (-1,\infty), W_m = \mathbb{R}$.



$\left[m \ \_\_\_\_, \quad y = \ln x \ ....., \right]$

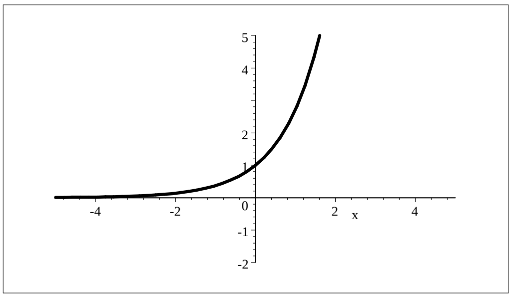
f. $n(x) = 1 + \ln x \qquad D_n = (0,\infty), \qquad W_n = \mathbb{R}.$

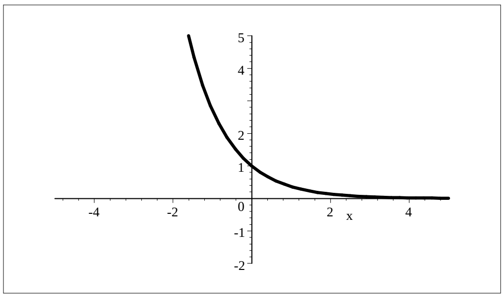


$\left[m \ \underline{\qquad}, \ y = 1 \ \ ..., \ y = \ln x \ \ ---\right]$

3.

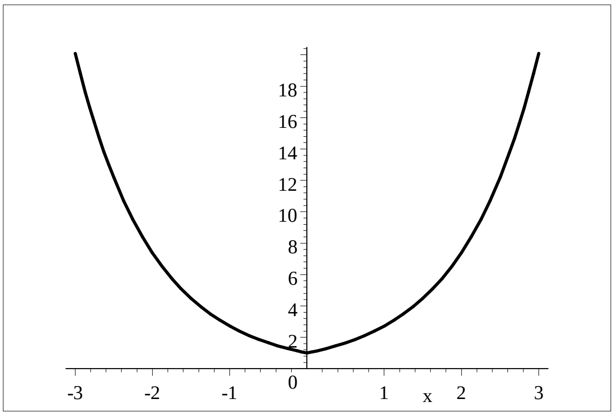
a. $f(x) = e^x, D_f = \mathbb{R}, W_f = [0,\infty).$ **b.** $(x) = e^{-x}, D_g = \mathbb{R}, W_g = [0,\infty).$



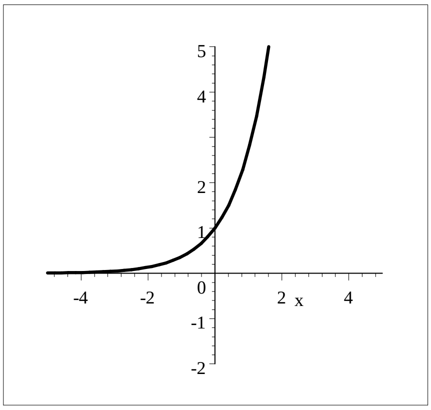


c. $(x) = e^{|x|} = \begin{cases} e^x & \text{if } x \geq 0 \\ e^{-x} & \text{if } x < 0 \end{cases}, D_h = \mathbb{R}, W_h = [1,\infty).$

d. $l(x) = |e^x| = e^x, \ \ D_l = \mathbb{R}, \ \ W_l = [0,\infty).$

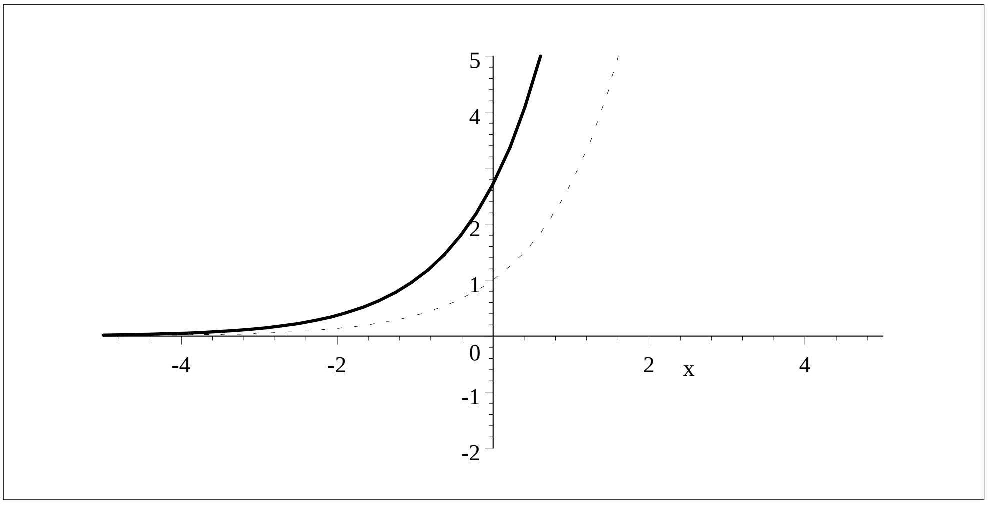


c.



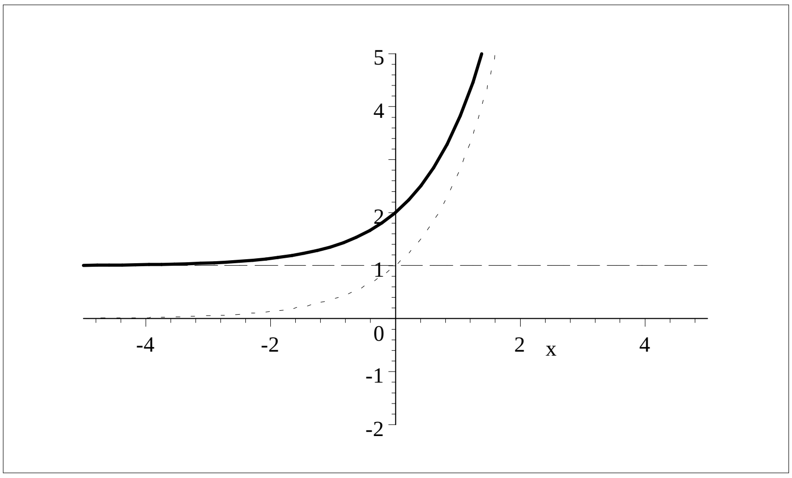
d.

e. $m(x) = e^{1+x}, D_m = \mathbb{R}, W_m = [0, \infty)$.



$\left[m \quad \_\_\_\_, \quad y = e^x \;\; ..... \right]$

f. $n(x) = 1 + e^x, D_n = \mathbb{R}, W_n = (1, \infty)$.



$\left[n \quad \_\_\_\_, \;\; y = 1 \quad ---, \;\; y = e^x \;\; ..... \right]$